```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>C&S Electric CSDPR-V2-500-A Relay Simulator</title>
    <style>
        :root {
            --bg-primary: #121417;
            --bg-secondary: #1a1d24;
            --panel-bg: #252a34;
            --accent-orange: #ff6f00;
            --lcd-blue: #a3ccff;
            --led-off: #3a1a1a;
            --led-on-red: #ff3333;
            --led-on-green: #33ff33;
            --text-color: #e0e6ed;
        }

        body {
            margin: 0;
            padding: 20px;
            background-color: var(--bg-primary);
            color: var(--text-color);
            font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
        }

        .dashboard {
            display: grid;
            grid-template-columns: 380px 1fr 420px;
            gap: 20px;
            max-width: 1600px;
            margin: 0 auto;
        }

        @media (max-width: 1200px) {
            .dashboard { grid-template-columns: 1fr; }
        }

        .card {
            background-color: var(--bg-secondary);
            border-radius: 8px;
            border: 1px solid #2d3341;
            padding: 15px;
```

```
    box-shadow: 0 4px 10px rgba(0,0,0,0.3);
}

h2 {
    margin-top: 0;
    font-size: 1.1rem;
    text-transform: uppercase;
    border-bottom: 2px solid var(--accent-orange);
    padding-bottom: 5px;
    color: #ffffff;
}

/* --- RELAY HARDWARE UI --- */
.relay-unit {
    background-color: #3b4252;
    border: 6px double #2e3440;
    border-radius: 12px;
    padding: 15px;
    color: #fff;
}

.brand-header {
    display: flex;
    justify-content: space-between;
    align-items: center;
    font-weight: bold;
    font-size: 1.2rem;
    color: #d8dee9;
    border-bottom: 2px dashed #4c566a;
    padding-bottom: 5px;
    margin-bottom: 10px;
}

.lcd-screen {
    background: #0f1c2e;
    border: 3px solid #1f232a;
    border-radius: 4px;
    height: 90px;
    padding: 10px;
    font-family: 'Courier New', Courier, monospace;
    color: var(--lcd-blue);
    font-size: 0.85rem;
    line-height: 1.3;
    overflow: hidden;
```

```
            box-shadow: inset 0 0 10px rgba(0,0,0,0.8);
            transition: all 0.3s;
        }
        .lcd-screen.backlight { background: #162f52; color: #c0ddff; text-shadow: 0 0 5px
rgba(163,204,255,0.6); }
        .lcd-screen.off { background: #05080c; color: #111; text-shadow: none; }

        .led-grid {
            display: grid;
            grid-template-columns: repeat(3, 1fr);
            gap: 12px;
            margin: 20px 0;
            background: rgba(0,0,0,0.15);
            padding: 10px;
            border-radius: 6px;
        }

        .led-node {
            display: flex;
            flex-direction: column;
            align-items: center;
            font-size: 0.7rem;
            font-weight: bold;
            text-align: center;
        }

        .led-dot {
            width: 14px;
            height: 14px;
            border-radius: 50%;
            background: #2a1111;
            margin-bottom: 4px;
            border: 1px solid #1a0808;
            box-shadow: inset 0 2px 4px rgba(0,0,0,0.6);
            transition: background 0.1s, box-shadow 0.1s;
        }
        .led-dot.red-on { background: var(--led-on-red); box-shadow: 0 0 8px var(--led-on-red); }
        .led-dot.green-on { background: var(--led-on-green); box-shadow: 0 0 8px
var(--led-on-green); }
        .led-dot.green-blink { animation: blink-g 1s infinite alternate; }
        @keyframes blink-g { 0% { background:#113a11; box-shadow:none;} 100% {
background:var(--led-on-green); box-shadow:0 0 8px var(--led-on-green);} }

        .hmi-buttons {
```

```
    display: grid;
    grid-template-columns: repeat(4, 1fr);
    gap: 8px;
}
.hmi-btn {
    background: #4c566a;
    border: 1px solid #2e3440;
    color: #fff;
    padding: 8px;
    border-radius: 4px;
    cursor: pointer;
    font-weight: bold;
    font-size: 0.8rem;
}
.hmi-btn:active { background: #3b4252; }

/* --- CONTROLS UI --- */
.control-group {
    margin-bottom: 12px;
}
label {
    display: block;
    font-size: 0.85rem;
    margin-bottom: 4px;
    color: #a0aec0;
}
select, input[type="range"], input[type="number"] {
    width: 100%;
    background: #2d3341;
    border: 1px solid #4a5568;
    color: #fff;
    padding: 6px;
    border-radius: 4px;
    box-sizing: border-box;
}

/* --- AR STATE MONITOR --- */
.ar-monitor {
    background: #11141a;
    padding: 10px;
    border-radius: 6px;
    margin-top: 15px;
    border-left: 4px solid var(--accent-orange);
}
```

```
.lockout-alert {
    background: #4a0e0e;
    color: #ff9999;
    padding: 10px;
    border-radius: 4px;
    text-align: center;
    font-weight: bold;
    border: 1px solid var(--led-on-red);
    animation: pulse-border 1s infinite alternate;
    display: none;
    margin-top: 10px;
}
@keyframes pulse-border { 0% { opacity:0.7; } 100% { opacity:1; } }

/* --- DATA LOGS & WAVEFORMS --- */
.table-wrapper {
    overflow-x: auto;
    margin-top: 10px;
}
table {
    width: 100%;
    border-collapse: collapse;
    font-size: 0.8rem;
    text-align: left;
}
th, td {
    padding: 8px;
    border-bottom: 1px solid #2d3341;
}
th { background-color: #1a1d24; color: #a0aec0; }

canvas {
    background: #0a0c10;
    border: 1px solid #2d3341;
    border-radius: 4px;
    width: 100%;
    height: 220px;
}
.metric-box {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 8px;
    margin-top: 8px;
    font-size: 0.75rem;
```

```
            background: #1a1d24;
            padding: 8px;
            border-radius: 4px;
        }
    </style>
</head>
<body>

<div class="dashboard">
    <!-- LEFT COLUMN: PHYSICAL RELAY & INJECTIONS -->
    <div style="display: flex; flex-direction: column; gap: 20px;">
        <div class="card relay-unit">
            <div class="brand-header">
                <span>C&S Electric</span>
                <span style="font-size: 0.75rem; color:
var(--accent-orange);">CSDPR-V2-500-A</span>
            </div>

            <div id="lcd" class="lcd-screen off">
                SYSTEM SHUTDOWN<br>NO POWER DETECTED
            </div>

            <div class="led-grid">
                <div class="led-node"><div id="led-l1" class="led-dot"></div>L1</div>
                <div class="led-node"><div id="led-l2" class="led-dot"></div>L2</div>
                <div class="led-node"><div id="led-l3" class="led-dot"></div>L3</div>
                <div class="led-node"><div id="led-e" class="led-dot"></div>E</div>
                <div class="led-node"><div id="led-pickup" class="led-dot"></div>PICKUP</div>
                <div class="led-node"><div id="led-trip" class="led-dot"></div>TRIP</div>
                <div class="led-node"><div id="led-block" class="led-dot"></div>BLOCK</div>
                <div class="led-node"><div id="led-cbfp" class="led-dot"></div>CBFP</div>
                <div class="led-node"><div id="led-ready" class="led-dot"></div>READY</div>
            </div>

            <div class="hmi-buttons">
                <button class="hmi-btn" onclick="pressHmi('<')">&lt;</button>
                <button class="hmi-btn" onclick="pressHmi('ENT')">ENTER</button>
                <button class="hmi-btn" onclick="pressHmi('>')">&gt;</button>
                <button class="hmi-btn" style="background: #8b0000;"
onclick="pressHmi('RST')">RESET</button>
            </div>
        </div>

        <div class="card">
```

```
        <h2>System Power & Injections</h2>
        <div class="control-group">
          <label>Source Matrix Configuration</label>
          <select id="power-source" onchange="updatePowerMatrix()">
            <option value="aux">Auxiliary Supply Active (Universal)</option>
            <option value="usb">Front USB Input Interface</option>
            <option value="battery">Internal Battery Module (3.6V)</option>
            <option value="none">No External Supply Connections</option>
          </select>
        </div>
        <div class="control-group">
          <label>Phase L1 Injection: <span id="val-l1">0.00</span> A</label>
          <input type="range" id="inj-l1" min="0" max="30" step="0.1" value="0"
oninput="processInjections()">
        </div>
        <div class="control-group">
          <label>Phase L2 Injection: <span id="val-l2">0.00</span> A</label>
          <input type="range" id="inj-l2" min="0" max="30" step="0.1" value="0"
oninput="processInjections()">
        </div>
        <div class="control-group">
          <label>Phase L3 Injection: <span id="val-l3">0.00</span> A</label>
          <input type="range" id="inj-l3" min="0" max="30" step="0.1" value="0"
oninput="processInjections()">
        </div>
        <div class="control-group">
```